2018 International Conference on Advances in Computing and Communication Engineering (ICACCE-2018)
Paris, France 22-23 June 2018

Enhancing the study of landslide prone area through supervised analysis: a case study of Varunavat Parvat, Uttarkashi, India

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Abstract— **Landslide occurrence in hilly regions of Uttarakhand has become a point of concern these days. Landslides are the most disaster causing activity happening in Gardhwal Himalayan terrain. It is becoming more regular threat over this region. This paper focuses on improving existing study using analytical techniques to understand the landslide occurrence situation clearly. The analytical approach may help identifying changing landslide patterns due to multiple landslide triggering factors. Further prediction methods can be used to predict future landslides. Present study can help government to take necessary actions preventing the monitory loss as well as human loss.**

Keywords--Triggering factors, Patterns, Predictive methods
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I. INTRODUCTION

Landslide is basically movement of soil, rocks, earth that occurs downward. A change in the stability of hilly regions are natural phenomena. The major triggering factors of landslides are heavy rainfall, earth vibrations, soil erosion, deforestation etc. The displacement of soil is considered the most dangerous and complex calamity happening on the earth surface. Figure 1 shows different forms of landslides. There is no defined standard and specific methodology to find out the next landslide. Slope failure is caused when gravitation force of earth that pulls the slope in downward direction increases the power of earth matter that slopes is composed of. Sliding of land can be very fast or it can be very slow both ways disturb the earth.

Landslide has become disastrous phenomena on the Varunavat hill of Uttarkashi district in Uttarakhand state [1]. It is causing damage to habitats, Property, and human lives every year. Uttarkashi is becoming prominent hotspot of landslide occurrences due to reasons like mountain building process, heavy rainfall, soil erosion, deforestation and many more.

Statistically landslides take more than 100 lives every year. Apart from this it damages infrastructure which cost in billions sometimes. Indirectly it is also harming natural environment, agriculture which takes several years to rebuild. These natural calamities cannot be prevented but human as well as economic loss can be minimized by taking appropriate measures. Therefore, identification of these landslide prone areas has become a priority these days for the safety of mankind. Methods and techniques such as heuristic, statistical, probabilistic and deterministic have been developed and applied for managing landslides. The statistical approach is found to be the most suitable for regional scale analysis to identify most frequent landslide zones with their causes. However, there is no standard method that has been accepted universally to identify patterns of landslides to predict them.

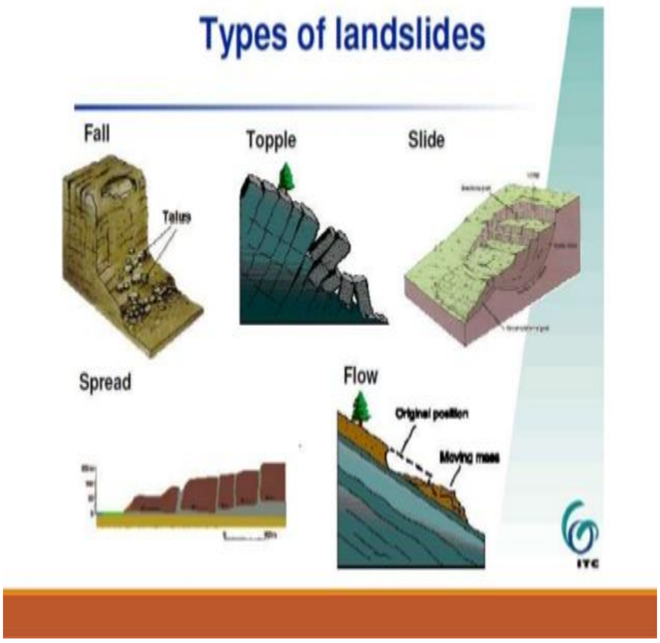

Fig.1 Types of Landslides

II. STUDY AREA

The 27th state of India is Uttarakhand which is famous for its beautiful Himalayas. Uttarakhand is situated in northern part of India and the area is 53,483 sq. km. The origin of all north India Rivers is Himalaya which is becoming victim of cloud bursts, landslide, and floods in last couple of years. After random interval of year's monsoon bring massive loss of lives, infrastructure, property, agriculture etc. Himalaya is facing natural calamities, resulting in people missing and becoming homeless. These disasters are not small to bear. Human life is suffering in different parts of the state. In hilly region of Uttarakhand there is no road connectivity so, it is difficult to save the loss of lives.

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Uttarkashi(long 78 degree 26' E, lat 30degree 44'N), a border district is very prone to natural disasters. Uttarkashi is situated on the base of Varunavat Parvat which is in the right side of river Bhagirathi at a height of 1150m.

Fig 2. Uttarkashi District on the Foothills of Varunavat Parvat

The earthquake in 1991 gave rise to unpredictable behavior in most of the areas of Uttarkashi. This includes Varunavat Parvat which was already prone to landslides from past decades.

On 24th Sept 2003, a highly destructive landslide took place on Varunavat Parvat. A multi-storied hotel was displaced from its location and many more buildings, road networks, infrastructure damaged in the sequence. The total damage calculated was about fifty millions dollars. This slide is considered as typical example of debris slide. Prior to this slide the Varunavat Parvat was affected by two landslides Gyansu and Tambakhani landslide [1].

Fig 3. Varunavat Parvat with fresh cracks

The Gyansu landslide took place in 1980(June). After that, the loose material made up of large rock and gravel regularly moves down constantly with the Gyansu nala. The landslide debris moves toward river Bhagirathi, blocking the Rishikesh–Gangotri route. Another landslide occurred was Tambakhani slide on varunavat parvat in September 2003 resulting the rolling boulders attacking multi-storied,in bus stand. Within 24 hours this slide converted into dangerous landslide burying hotels, commercial buildings, houses, roads etc.

Above 50 meters and on east side of Tambakhani slide there is scarp which occurs with earlier slides. Cracks were also seen in EW direction after these landslides. The study [2] shows that the hill was not stable before the 24th September 2003 landslide. However, Varunavat hill is prone to slow moving landslide which was not affecting social life, but threat is on today as well. These slow moving landslides leads disastrous landslide harming life and land.

The consequences of these disasters are human loss, habitat loss, and unavailibity of drinking water supply, improper arrangement of sanitation, poor food service, and lack of rescue services. Those who are saved somehow, die by mismanagement.

Fig 4. Landslide scenario in Varunavat Parvat

III. Literature Review

The literature review shows that several studies has been done for study area but technical approach has not been discussed for future forecasting of landslide.

The study of The 23 September 2003 Varunavat Parvat landslide in Uttarkashi township, Uttaranchal was done by Vikram Gupta et al. in 2004 . In this study the approach was to displace the material using volume calculation formula based on ellipsoid geometry of the landslide mass leading short term mitigation method[2].

Later in the same year S.Sarkar et al. done risk analysis for the study area after the landslide disaster of 24th September 2003[3]. Analysis of the 2003 Varunawat Landslide, Uttarkashi, India using Earth Observation data was done by K. Vinod kumar in 2007. High-resolution stereoscopic earth observation data was used to study landslide in detail[4]. In 2008 A case study of Varunavat Parvat landslide in Uttarkashi township was presented by Anirudh Uniyal where mitigation strategy was discussed for landslide prone areas[5]. The methodology approach was monitoring landslide zones via survey, satellite images, and interaction with villagers. Also in 2008 probable causes of landslides were identified and stability analysis is done by R.S.T .Sai et al[6]. In 2009 stabilization of varunavat landslide study was done by Dr. P.C.Nawani for which approached methodology was surveys, geological mappings , slope stability analysis using numerical modeling and monitoring of Automatic target recognition by WHIG[7].

S.Sarkar et al. in 2010 presented study where causative factors of landslides and associated risks were described[8].

In 2015 Pramod C. Nawani presented paper for varunavat parvat landslide treatment . The approach was to evaluate the performance of supports provided in the slope . Stability analysis is done using numerical modeling[9].

IV. RESEARCH GAP

Landslide in Varunavat parvat is very old and continuous phenomenon. Many successful studies has been from decades to understand the causative factors of landslides, stability analysis of slope. But still there is need of system which can analyze the triggering factors of study area and predict future landslides as alert. To fill this gap few analytical and prediction techniques are discussed in the paper to achieve the goal.

V. METHODOLOGY

Supervised learning of the study area can be achieved by following steps

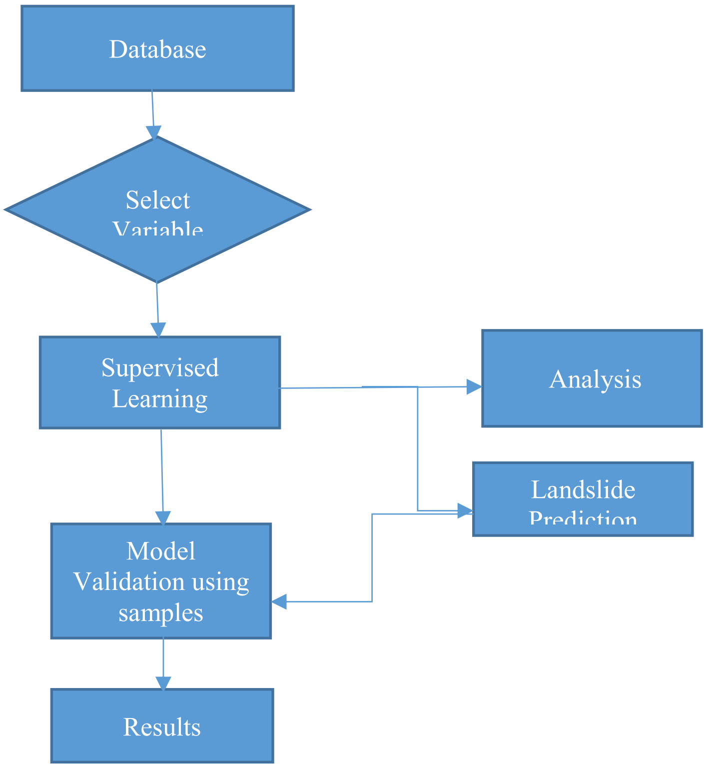

Fig 5. Proposed Methodology Flowchart

A. *Data preparation*

Data collection and preparation for selected area is a sensitive task. To study the area in detail and finding out the triggering factors and combination of factors need more attention while preparing database. The learning can be more effective when all the factors associated to landslides are collected. In this research landslide data can be in the form of geospatial data which will include polygons, lines and points with associated latitude and longitudes. There can also be satellite images, surveys. The landslide inventory data are assumed to be most critical factor for predicting future landslides. There are different types of landslides such as fall, topple, rotational slide, spread, flow. Triggering factors like Slope Instability, groundwater, geological structure, rainfall, nature of soil, aspect, and distance from river, distance from road, land cover, lithology, drainage line, pro-curvature, residential area, geological conditions are considered in the analysis of landslide occurrence. Some major factors are discussed in the paper.

B. *Slope Instability*

Instability of slope in Varunavat parvat is one of the major cause of landslide. The instability existed when two major landslides Gyansu Slide , June 1980 and Tambakhani slide occurred. Since then the loose material and debris is continuously flowing towards downward direction. Later, earthquake in Uttarkashi made slope instable and caused many cracks in the hill [1].

C. *Rainfall*

According to study some instability is already existing in Varunavat parvat , landslide occurs when heavy rainfall is there loosening the mud and bringing it in foot of the hill. The heavy rainfall on 20-23 September 2003 induced pore pressure which re-activated the old slide. This result in loss of property, habitats and national highways [2].

D. *Data analysis and Prediction models*

In this research supervised learning can be achieved using linear regression model, Artificial Neural network and Support Vector machine. These models are used to find the relation between landslide triggering factors. Later, the performance of all the models for same dataset will be assessed using SPSS.

1. *Artificial Neural Network*

Artificial Neural Networks was introduced in 1943 by Warren McCulloch, a neurophysiologist, and a young mathematician, Walter Pitts. They modelled network of neurons using electrical circuits explaining how neurons actually works. An artificial neural network is a "computational mechanism able to acquire, represent, and compute a mapping from one multivariate space of information to another, given a set of data representing that mapping" (Garrett, 1994).Landslides in hilly regions like varunavat parvat in Uttarakashi , Uttrakhand are very prone to landslides. An effective solution is required to analyze and predict future landslides. ANN is an effective solution which can sense the relationship between multiple factors. ANN supports both qualitative data and quantitative data [10].

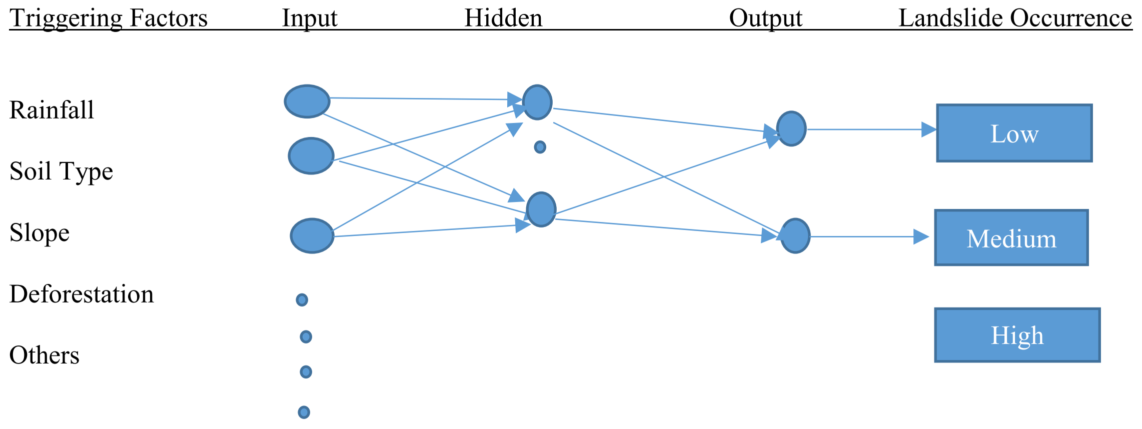

Fig 6.Artificial Neural Network Model

2. Support Vector Machine

SVM is a supervised leaning method which analyze data used for classification and analysis.
Given training dataset of n points such as

$$(\vec{x}_1, y_1), \ldots, (\vec{x}_n, y_n)$$

Where, y_i can be 1 or -1 indicating to class where $\vec{x}_i$ belongs
The idea is find out the optimal hyperplane which separate classes efficiently.
Hyperplane can be defined as

$$\vec{w} \cdot \vec{x} - b = 0,$$

Where, $\vec{w}$ is a normal vector to hyperplane

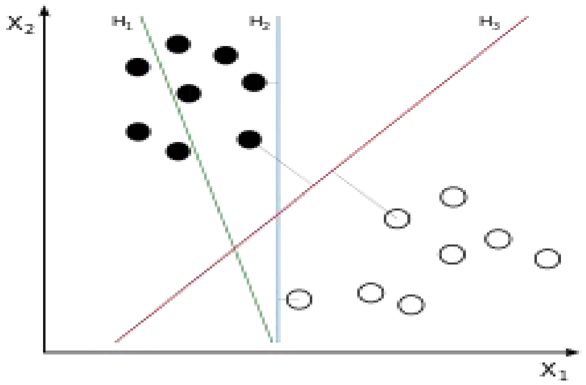

Fig7. Support Vector Machine

3. Logistic Regression

Logistic regression is an analytical method to analyze set of data in which there are more than one independent variables which determines result which is dichotomous. This method uses more than one predictor parameter continuous or categorical. Types of logistic regression: binomial (binary logistic regression), ordinal or multinomial. Binomial is useful when observed result for a dependent parameter can have two value 0 or 1. Second method is useful when dependent variables are in some order. And last one is useful when result can have more than two values.
Standard Logistic function is defined as

$$\sigma(t) = \frac{e^t}{e^t + 1} = \frac{1}{1 + e^{-t}}$$

Where, t is a real input
$(t \in \mathbb{R})$, And $\sigma(t) \in (0, 1)$ for all t.

In context to landslide prediction all the triggering factors of landslide will be independent variables and the outcome YES or NO (presence or absence of landslide) will be dependent variable.
Logistic formula defined in terms of probability is Logit (p) =

$$\ln\left(\frac{\hat{p}}{1 - \hat{p}}\right) = B_0 + B_1 X$$

Where, ln stands for natural logarithm and $B_0 + B_1 X$ is a line of regression.

E. *Model Validation*

To validate proposed models training dataset testing samples from database will be generated. The training dataset will be used for supervised learning on different models. Sample data will be used to validate the performance of proposed models and calculating the accuracy of models.

TABLE 1: COMPARISON STUDY OF ANALYZING AND PREDICTING LANDSLIDES.

Following table shows the comparison chart of proposed methods in terms of performance in analyzing and predicting landslides.

S.No	Author	Year	Paper Title	Methods	Conclusion
1	Bahareh Kalander et al.	2017	Assessment of the effects of training data selection on the landslide susceptibility mapping: a comparison between support vector machine (SVM), logistic regression (LR) and artificial neural networks (ANN)	AVM, ANN, Logistic Regression	Overall Accuracy SVM=79.82% LR=81.42% ANN=70.2%
2	LimYi Zhou et al.	2017	A comparative study of slope failure prediction using logistic regression, support vector machine and least square support vector machine models	Logistic regress ,SVM, Least Square SVM	1st case-Performance is similar 2nd case-LR has best predictive efficiency
3	Haoyuan Hong et al.	2017	Providing a Landslide Susceptibility Map in Nancheng County, China, by Implementing Support Vector Machines	SVM, Naïve Bayes Factors-: lithology, soil, slope, aspect, elevation, topographic wetness index, distance to rivers and distance to faults	Accuracy SVM-89.70% Naïve Bayes-86.78%
4	Liang-Jie Wang et al.	2016	A comparative study of landslide susceptibility maps using logistic regression, frequency ratio, decision tree, weights of evidence and artificial neural network	Logistic Regression, frequency ratio, decision tree, weight of evidence, ANN	Logistic Regression performed best with highest AUC(.865)
5	Neenu Rachel et al.	2016	Landslide Prediction with Rainfall Analysis using Support Vector Machine	SVM, ANN	SVM performed best
6	Cristiano Ballabio et al.	2012	Support Vector Machines for Landslide Susceptibility Mapping: The Staffora River Basin Case Study, Italy	SVM, Logistic Regression, LDA, Naïve Baiyes	SVM perform best in accuracy. Also SVM is less prone to false positives

VI. DISCUSSION AND CONCLUSION

In landslide management factor analysis and future forecasting of landslides can help to prevent and manage situation effectively. In this paper a model is proposed for analyzing landslide occurrence in varunavat parvat of Uttarkashi in Uttarakhand. Taking triggering factors such as Slope Instability, groundwater, geological structure, rainfall, nature of soil, aspect, and distance from river, distance from road, land cover, lithology, drainage line, pro-curvature, residential area and geological conditions into consideration methods such as SVM , ANN and logistic regression will be used to analyze the data and to predict future landslide. The outcomes of prediction will be divided into classes(LOW/MEDIUM, or YES/NO). The comparison table shows that SVM perform well in terms of accuracy for considered factors. In some case logistic regression is performing better than SVM but SVM model best fits the hyperplane which divide the two groups landslide and No-landslide efficiently. One more advantage of using SVM is it works well with data having high dimension. Although SVM accuracy depends on landslide conditioning factors as they vary according to geographic locations.

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